ABSOLUTE VALUE AS PIECEWISE

Removing the bars

A transformed absolute value is two linear formulas, one on each side of the vertex.

01

ABSOLUTE VALUE FORM — BEFORE

$$f(x) = |x - 3| + 2$$

↓ Split at the vertex — |x − 3| is x − 3 for x ≥ 3 and −(x − 3) for x < 3

02

TWO PIECES, UNSIMPLIFIED

$$\begin{cases} (x - 3) + 2 & \text{if } x \geq 3 \\ -(x - 3) + 2 & \text{if } x < 3 \end{cases}$$

↓ Simplify each piece — each one is an ordinary linear formula

03

PIECEWISE FORM — AFTER

$$\begin{cases} x - 1 & \text{if } x \geq 3 \\ -x + 5 & \text{if } x < 3 \end{cases}$$

▼ WHERE THEY MEET